

Student Dropout Risk Analysis Dashboard

OULAD Dataset | Predicting Student Success & Withdrawal

Total Students

28,785

Average Score %

72.8

Dropout Ratio %

31.6

Total Dropout

9,082

High Risk Students

951

Key insights

Risk:
31.2% of students withdrew — highest actionable segment to target for early intervention.

Gender:
Male students account for 54.8% of dropouts; gender-specific support programs may help close the gap.

Module:
FFF & BBB modules drive the most withdrawals — curriculum review or added tutoring recommended.

Risk flag:
951 high-risk students are currently active — proactive outreach could recover a significant portion.

Strength:
72.8 average score and 9.3% distinction rate show a healthy top tier worth nurturing.

Region

All

Highest_Educa...

All

Age_band

All

Gender

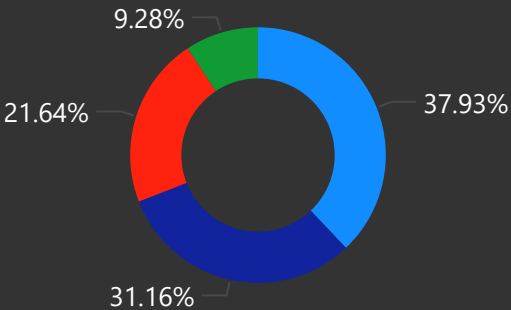
☒ F
☒ M

Disability

☒ N
☒ Y

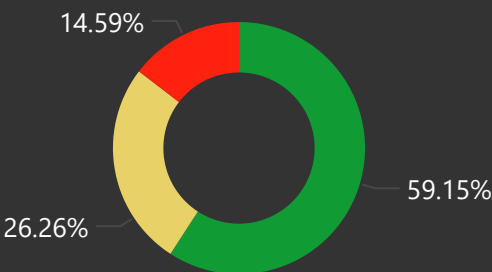
Final Result Distribution

Pass Withdrawn Fail Distinction



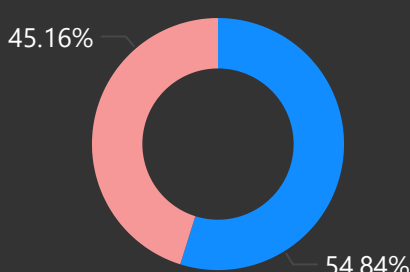
Risk Distribution

Low Risk Medium Risk High Risk

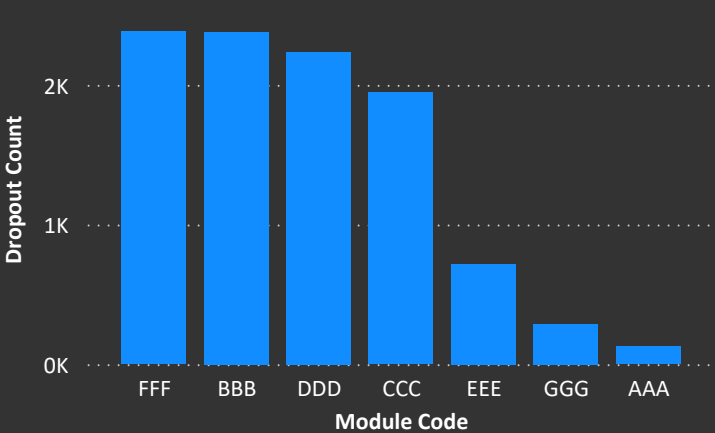


Dropout Rate by Gender

M F



Top Modules with Highest Dropout



Risk Summary

Risk_Level	Students	Pass Rate %	Avg Clicks	Dropout Rate %
High Risk	951	4.47	1,503.83	89.04
Medium Risk	1712	9.34	6,193.80	59.73
Low Risk	3856	65.55	15,487.71	11.09
Total	6519	41.27	95,519.68	31.55

Student Engagement Analysis

Student Engagement & Performance

Avg VLE Clicks

1.36K

Avg Interection

364.56

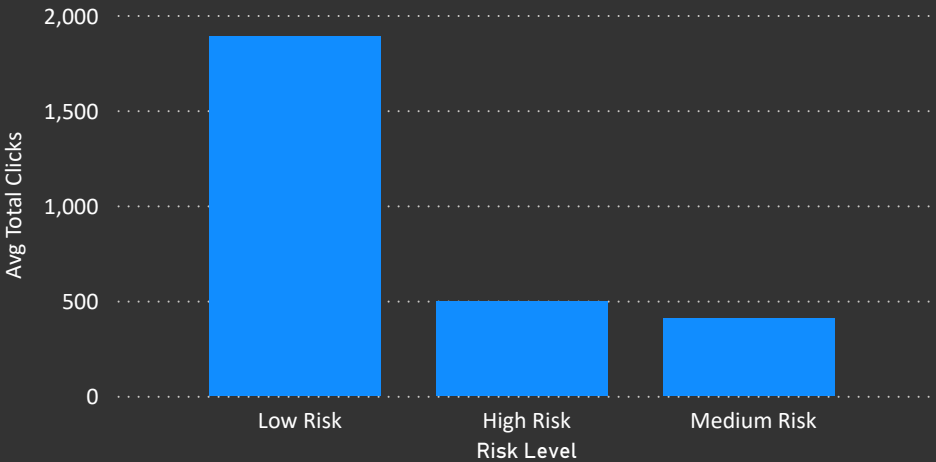
Avg Engagement Score

3.22

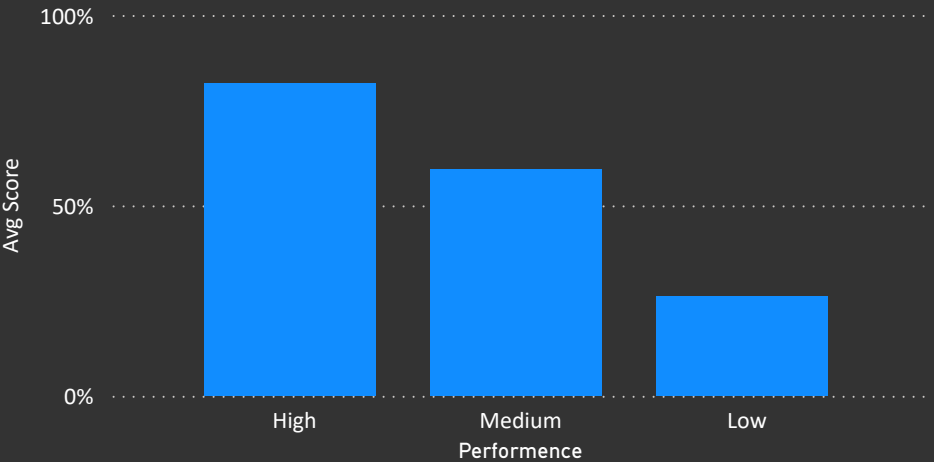
Avg Study Days

407.19

Average Clicks by Risk Level



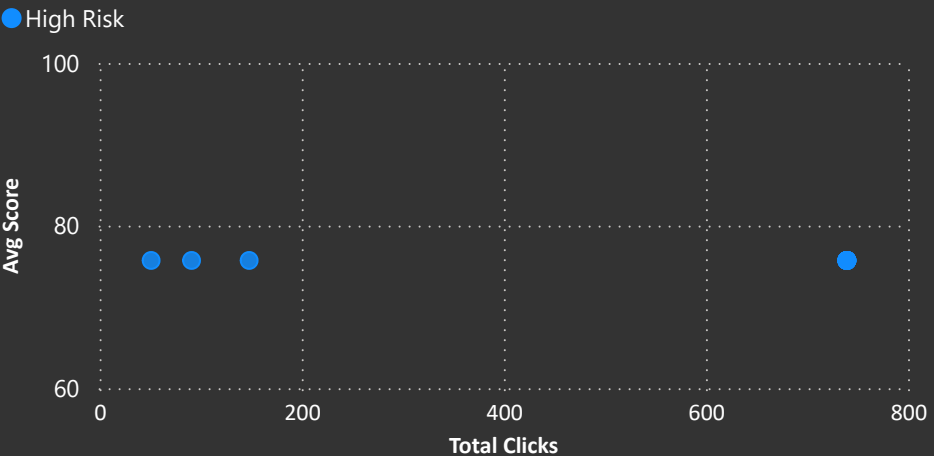
Average Score by Risk Level



Top Engaged Students

Student ID	Total_Clicks	Avg_Score	Risk_Level
80868	28615	150.33	Low Risk
537811	25159	291.21	Low Risk
557710	15300	90.25	Low Risk
597365	16379	173.38	Low Risk
599577	15931	89.00	Low Risk
624060	16980	198.00	Low Risk
630905	28264	192.67	Low Risk
680027	16153	196.86	Low Risk
1576513	15716	86.60	Low Risk
2063578	17957	186.96	Low Risk
Total	196454	1,655.25	

Engagement VS Performance



Key insights

Clicks predict risk with high accuracy

- Low-risk students click nearly 4x more than high-risk students (~1,900 vs ~530). Click rate alone is a reliable early-warning signal.

Score gap mirrors engagement gap

- Low-risk students perform ~75–80% on average vs ~45–50% for high-risk — both metrics drop together, reinforcing disengagement as the root cause.

Top engaged students are all low-risk

- Every student in the top-10 clicks table is low-risk, confirming that high engagement is a strong protective factor against dropout.

Engagement drop = actionable trigger

- A sudden weekly drop in VLE clicks should trigger proactive outreach before the student fully disengages.

Dropout Prediction Model Results

Accuracy
85%

Precision
78.0%

Recall
70%

F1 Score
74%

ROC AUC
91%

Key insights

Model ranks risk very well

- ROC AUC of 0.92 shows strong discrimination between dropout and non-dropout students across all thresholds.

611 real dropouts are being missed

- Recall of 70% means 611 actual dropouts go undetected. These are the highest-priority students for manual review or rule-based alerts.

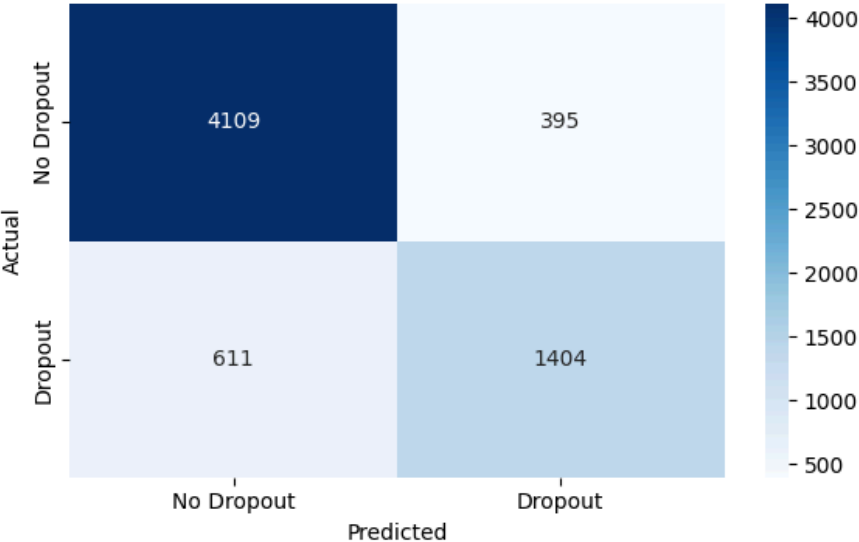
Threshold tuning is the key lever

- Lowering the decision threshold will catch more dropouts at the cost of more false alarms (395 today) — a worthwhile trade-off for education.

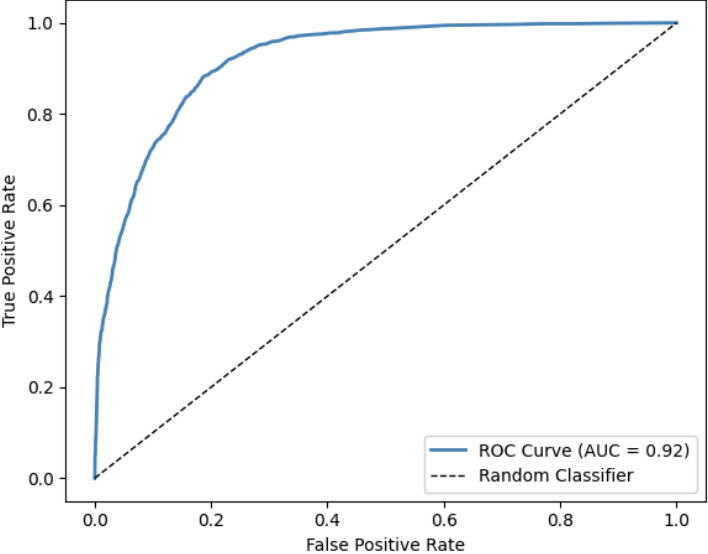
False alarms are manageable

- 78% precision means 1 in 5 flagged students won't drop out — acceptable given intervention is low-cost relative to losing a student.

Confusion Matrix



ROC Curve — Random Forest



Top High-Risk Students Table

Student ID	Dropout Probability	Risk Level
339353	1.90	High Risk
675116	1.90	High Risk
612805	1.88	High Risk
599547	1.82	High Risk
639232	1.82	High Risk
677721	1.72	High Risk
528642	1.66	High Risk
603053	1.66	High Risk
Total	812.82	

Precision, Recall & F1 Score by Class

